Function

- Calcium is one of the minerals needed for the growth and strength of the bones.
- Calcium is needed for muscle contraction
- Calcium ions are essential for blood clotting.
- Calcium is needed for the functioning of neuro transmitters.

Food sources

Milk and milk products are among the best sources of calcium, other sources of calcium includes leafy greens, firm tofu, and small fish with bones, such as sardines.

Deficiency

A severe deficiency of calcium leads to the condition hypocalcemia resulting in rickets in children and osteomalacia in adults.

Phosphorus

Approximately 85% of the 700 grams of phosphorus in the adult body is present in the bones. The ratio of calcium to phosphorus in the bones is 2: 1. After calcium, phosphorus is the second most common mineral in the body by weight.

Function

- Phosphorus is involved in a variety of chemical reactions in the body, many of which are related to energy metabolism.
- Mineralization of bones and teeth
- Facilitation of energy transaction
- Absorption and transport of nutrients
- Regulation of protein activity
- Component of essential body compounds.

Food sources

In general, good sources of protein are also good sources of phosphorus. Meat, poultry, fish and eggs are rich in phosphorus. Milk and milk products, nuts, legumes, cereals and grams are good sources.

Deficiency

A phosphorus deficiency is characterized by weakness, lack of appetite, fatigue and muscle pain.

Magnesium

About 60% of the body's magnesium is contained in the bones; most of the remaining magnesium is present in the muscles and other tissues, with about 1% circulating extracellular fluids.

Function

- The activity of hundreds of enzymes depends on magnesium
- Magnesium also helps maintain calcium and potassium homeostasis.

Food sources

Nuts, legumes, dark green leafy vegetables, soya beans and milk are good sources.

Deficiency

Symptoms of magnesium deficiency include weakness, confusion, lack of appetite, nausea, and lack of coordination.

Sulfur

Because sulfur is part of the essential amino acid methionine and the nonessential amino acid cysteine, it is present in the body's proteins. Interactions among sulfur atoms in cysteine helps to give proteins their three dimensional shape.

The Electrolytes

Three of the macrominerals – Sodium, potassium and chloride are the body's main electrolytes. The major function of the electrolytes is to maintain the proper distribution of water inside and outside cells. If this water balance is not maintained, cells will shrink or swell beyond their normal size. Sodium and chloride are found in the extracellular fluid; potassium is found in the intracellular Fluid (ICF).

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